

“ENFORCEMENT AS MESSAGE”
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THE PAPER

Main conceptual point: Compliance with the law depends on the perceived commitment of the state to the law.

Therefore, the level of enforcement not only reduces violations directly, but also signals commitment to the law.

THE MODEL

The state is **committed** ($\theta = C$) or **strategic** ($\theta = S$).

If $\theta = C$, the state tries to punish every violation.

If $\theta = S$, the state allows violations $v \leq \Delta$ in order to minimize enforcement costs.

Δ is the level of forbearance, inverse of the level of enforcement.

Legitimacy is μ , the (common) belief among citizens that $\theta = C$.

Citizens with type $\gamma \leq \mu$ don't violate ($v = 0$). This is **voluntary compliance**.

Those with $\gamma > \mu$ choose v strategically to maximize

$$b(v) - [\mu + (1 - \mu)1_{v > \Delta}]wp(v).$$

The solution is $v^*(\mu, \Delta) > 0$.

We have $\gamma \sim F$. The *aggregate violation level* is $X(\mu, \Delta) = (1 - F(\mu))v^*(\mu, \Delta)$.

THE MODEL

At first θ is drawn with $\Pr(\theta = C) = \mu_0$.

Then at each $t = 0, 1, \dots$,

1. The state chooses $\Delta_t \geq 0$, and the citizens γ (simultaneously) choose $v_t(\gamma) \geq 0$.
2. Everybody observes a signal $\mathbf{s}_t = \Delta_t + \epsilon_t$, where $\epsilon_t \sim G$ iid.

The strategic type minimizes discounted enforcement costs $\sum_{t \geq 0} \beta^t \mathbf{K}(\mathbf{X}_t, \Delta_t)$.

We assume $K_X > 0$ (violations are bad) but $K_\Delta < 0$ (forbearance saves costs).

The equilibrium concept is MPBE. The payoff-relevant state is μ_t .

EQUILIBRIUM

Let $V(\mu)$ be the expected discounted cost for the strategic type if $\mu_0 = \mu$.

We have the recursion

$$V(\mu) = K(X(\mu, \Delta), \Delta) + \beta \int V(\mu'(\mu, s, \Delta))g(s - \Delta) ds,$$

where

$$\mu'(\mu, s, \Delta) = \frac{\mu g(s)}{\mu g(s) + (1 - \mu)g(s - \Delta)}$$

is the posterior if we expect S to do Δ and we observe signal s , and

$$\Delta(\mu) \in \arg \min_{\sigma} \left\{ K(X(\mu, \Delta), \sigma) + \beta \int V(\mu'(\mu, s, \Delta))g(s - \sigma) ds \right\}$$

is the optimal forbearance, taking into account that in equilibrium it coincides with the citizens' expectation.

Result. There is an equilibrium $V(\mu)$ and a policy $\Delta(\mu)$ that implements it.

MAIN RESULTS

If $\beta = 0$ (the future doesn't matter) then the strategic type doesn't attempt to enforce the law ($\Delta = 1 - r$, the maximal level), because it cannot commit to do so ($K_\Delta < 0$). The citizens expect this.

If $\beta > 0$ then the strategic type doesn't fully enforce ($\Delta_t > 0$).

[The paper doesn't show that the strategic type chooses to enforce at least a little bit ($\Delta_t < 1 - r$) when, say, $\beta > \underline{\beta}$ for some $\underline{\beta} \in [0, 1)$. But I think that it should show sufficient conditions under which this is the case.]

If the state is strategic we have $\mu_t \rightarrow 0$ almost surely as $t \rightarrow +\infty$, i.e., **the state gradually loses legitimacy**.

If the state is committed then $\mu_t \rightarrow 1$ almost surely as $t \rightarrow +\infty$, and **voluntary compliance becomes universal**.

COMMENT 1

What happens to the forbearance level Δ on the equilibrium path?

One intuition is that enforcement and legitimacy are **complements**: if μ is low then there is no point in using enforcement to pretend to be committed.

(Also $K_{X\Delta} < 0$ by assumption: $\mu \uparrow \Rightarrow X \downarrow \Rightarrow \Delta \downarrow$ through this channel, so enforcement $\uparrow$.)

Another intuition is that enforcement and legitimacy are **substitutes**: if μ is high then there is no point in enforcing, because very few people are going to violate, and people are not going to pay much attention to the signal s anyway.

It would be great to prove that Δ is U shaped in μ , maybe under stronger assumptions.

COMMENT 2

Citizens don't observe if they are punished. How to defend this assumption?

If citizens observed their own punishment (totally plausible), they would learn about θ . But then each citizen would have a different belief, which complicates things significantly.

Also, enforcement Δ doesn't directly affect the aggregate violation level X . This means: if you police more you don't reduce crime, because by the time criminals learn that you did they already committed the crime. I understand why this is assumed, but it's a bit counterintuitive.

COMMENT 3

The paper starts with the question “Why do some legal rules become social norms while others remain merely formal commands?”

Answer: the law becomes a social norm iff the state is committed to it. If the state is not committed, eventually people comply only because of enforcement.

How this works: by not tolerating violations the committed state **non-strategically** communicates that it is committed.

A thought: suppose that the committed state was also strategic, but it cared about compliance more than the strategic type. (E.g., the committed type wants crime to be lower everywhere, but the strategic type only cares about electorally pivotal districts, say.)

The Acemoglu, Egorov and Sonin (2013) logic could kick in: the committed type could “over-enforce” relative to the optimum in order to signal that it is committed. The dynamics could be interesting.

COMMENT 4

Is this actually about “social norms”?

Maybe not. The legitimacy is in the relationship between society and the state, not in the relationship between society and itself (which the term “social norms” implies).

In the literature on norms a common interpretation is that norm-following derives from a preference for coordination. The “social norm” is whatever people expect that others do.

In this paper voluntary compliance happens iff the perceived commitment of the state is at least as large as the intrinsic commitment of the citizen. It’s a sort of desire to coordinate motivations (not actions per se).

COMMENT 5: WHAT ABOUT WELFARE?

- One interpretation is that $K(X, \Delta)$ captures the full social value of the law.

What the model maybe shows is that a “rational” (i.e., strategic) state does (or can do) less well than a committed state. It would be nice to show this explicitly.

What’s interesting is that in this case the paper provides an “irrational” (but functional) explanation for “legal rationality.”

- Another route is to assume that only the committed type internalizes the full social value of the law, and with that objective full enforcement ($\Delta = 0$) is optimal. In that case the “strategic” type is bad.
- Another route is to connect the citizens’ payoffs to social welfare. And maybe justify the need for a law from an externality, say. The paper can ask: is a legitimate law always good? Maybe $\mu = 0$ can be good under some circumstances.

COMMENT 6

Maybe it's possible to simplify the model, and this may help prove more results?

Aggregate violations $X(\mu, \Delta) = (1 - F(\mu))v^*(\mu, \Delta)$ can be non-monotonic in both arguments, I think.

But intuitively more legitimacy should reduce violations ($X_\mu < 0$) and more forbearance should increase violations ($X_\Delta > 0$).

For example, suppose that $v \in \{0, 1\}$. Assume that if $\gamma \leq \mu$ then $v = 0$, and if $\gamma > \mu$ then $v = 1$ unless the citizen is detected. $\Delta \in [0, 1]$ is the fraction of citizens who are not monitored. Then $X = (1 - F(\mu))\Delta$.

CONCLUDING COMMENTS

New perspective on an old question, making a connection with the philosophy of law (something valuable in my opinion).

Technically impeccable.

It provides a way of modeling legal legitimacy that allows us to ask other interesting questions, e.g., about the relationship between legal legitimacy and economic efficiency, or between legal legitimacy and political institutions.

You should read it when it's ready!